Chiral generating pairs in finite simple groups: a complete assembly of the positive theorem

Human Compression project

Abstract

We assemble a proof of the positive existence theorem of Leemans and Liebeck for finite nonabelian simple groups. The proof combines explicit alternating generators, finite word certificates, and involution-counting arguments with all parameter ranges accounted for. The classification and geometric reconstruction theorems used as antecedents are stated explicitly. Several auxiliary assertions in the original proof are replaced by corrected arguments, without changing its positive conclusion. No new existence theorem or classification is claimed.

The full proofs and certificates used below are retained in their exact component editions. The accepted general orthogonal counting argument and the separately accepted D_4^+ cover are joined explicitly in Section 4.4. Historical pending-review language in a frozen component records its earlier author stage; the acceptance and dependency records bound to this assembly state the completed interfaces. This manuscript establishes the positive principal theorem, with the source boundaries specified in Section 5.

1 The exact theorem and its geometric meaning

The fixed source is arXiv:1603.07713v3, whose running header is dated 18 May 2016 and whose internal date is 6 November 2018. Its numbered principal theorem is the following positive statement [1].

Theorem 1 (Leemans–Liebeck)**.** *Let G be a finite nonabelian simple group which is not isomorphic to A_7, $\mathrm{PSL}_2(q)$, $\mathrm{PSL}_3(q)$ or $\mathrm{PSU}_3(q)$. There are $x, t \in G$ such that*

$$G = \langle x, t\rangle, \qquad |t| = 2, \qquad \nexists \alpha \in \mathrm{Aut}(G): \quad x^\alpha = x^{-1}, \quad t^\alpha = t.$$

The source asks only that no *involutory* automorphism have the specified action. These formulations are equivalent for a generating pair: the square of such an automorphism fixes both generators and hence is the identity. The identity itself cannot invert x, since $|x| > 2$. Indeed, a group generated by two involutions is dihedral, whereas G is nonabelian simple. The automorphism group in the theorem is the full $\mathrm{Aut}(G)$; restricting to inner automorphisms would not suffice.

Lemma 2. *Every pair in Theorem 1 satisfies the rank-three rotation intersection conditions and yields an abstract chiral polyhedron with full automorphism group G.*

Proof. Set $y = x^{-1}t$. Then $xy = t$, so $(xy)^2 = 1$, and $G = \langle x, y\rangle$. An element of $\langle x\rangle \cap \langle y\rangle$ commutes with both generators, and therefore is central in G. This intersection is trivial. Neither x nor y has order at most two: otherwise G would be cyclic or generated by two involutions. Thus the orders are at least three.

We now invoke the retained Schulte–Weiss rank-three reconstruction and regularity criterion [2], in the form used in the source: these rotation data reconstruct a directly regular or chiral polyhedron; the directly regular alternative is precisely the existence of the reversing automorphism in Theorem 1. Its absence gives chirality, and the reconstructed full automorphism group is G. □

Corollary 3. *Every group in Theorem 1 is the full automorphism group of an abstract chiral polyhedron.*

This uses geometric reconstruction as an explicit antecedent, rather than claiming a new construction theorem for abstract polytopes. Equivalence of marked rotation data means a group isomorphism carrying the distinguished generators to their counterparts. No bijection between *literal labelled pairs* and unmarked polyhedra is asserted here. In particular, no enumerative count is deduced by forgetting orientation, identifying enantiomorphs, or dividing the number of pairs by an unstated symmetry factor.

2 Two sufficient criteria

The following elementary arguments explain the two forms of certificate used throughout the proof. Write $i_2(H)$ for the number of elements of order exactly two, and $\nu_2(H) = i_2(H) + 1$ for the number of solutions of $h^2 = 1$.

Lemma 4 (Word criterion)**.** *If $G = \langle x, t\rangle$, $|t| = 2$, and a word w satisfies $|w(x,t)| \neq |w(x^{-1},t)|$, no automorphism of G inverts x and fixes t.*

Proof. An automorphism with those generator images sends one word value to the other, whereas automorphisms preserve element order. □

All forty-seven explicit finite certificates used below employ $w(x,t) = xtx^2tx^3t$: twelve sporadic cases and thirty-five Lie parameter entries. These are parameter certificates, not forty-seven distinct abstract groups, since some small classical names coincide.

Lemma 5 (Counting criterion)**.** *Let G be finite and centreless, and fix $x \in G$ of order greater than two. Suppose $\mathcal{H}$ is a family of actual proper subgroups covering every proper subgroup containing x. If*

$$i_2(G) > \sum_{\substack{\alpha \in \mathrm{Aut}(G) \\ |\alpha|=2,\ x^\alpha = x^{-1}}} i_2(C_G(\alpha)) + \sum_{H \in \mathcal{H}} i_2(H),$$

then an involution t satisfies the conclusion of Theorem 1.

Proof. The second sum bounds involutions that generate a proper subgroup with x. The first bounds involutions fixed by any involutory reverser. Choose t outside their union. Any reverser fixing this generating partner has square one, as observed above, and is already excluded. □

Overlaps among these sets only enlarge the bound, so no disjointness is required. A list of *types* or conjugacy classes of subgroups is not yet $\mathcal{H}$: the number of actual subgroups containing x must be proved. The cyclic normalizer must also be covered or separately charged. These points are paid explicitly by the acquired components.

A convenient bound for the first sum is the following. If $G \leq J \lhd \mathrm{Aut}(G)$ and $C_J(x) = T$, each outer coset contains at most $|T|$ reversers. Only square-one elements of $\mathrm{Aut}(G)/J$ can be images of involutory reversers. If their number is κ and every relevant fixed group has at most μ involutions, the first sum is at most $\kappa|T|\mu$. This does not assume that all reversers are conjugate. When passing through a kernel K, we use $\nu_2(H) \leq |K|\nu_2(H/K)$, including the identity before taking fibres. No algebraic dimension estimate is substituted for a finite kernel order.

3 Alternating and sporadic groups

For A_n, $n \geq 8$, the complete alternating component [3] proves generation and excludes the full reversing automorphism group for the source's explicit pairs:

n even	$x = (2,3,\ldots,n)$	$t = (1,2)(3,4)$,
n odd	$x = (1,2,\ldots,n)$	$t = (1,2)(3,6)$.

Its proof is uniform in n; the finite controls only check displayed identities and do not prove an unbounded theorem by sampling. The standard alternating-group automorphism theorem and the stated primitive permutation-group antecedent remain retained. The lower alternating names $A_5 \cong \mathrm{PSL}_2(4)$ and $A_6 \cong \mathrm{PSL}_2(9)$ lie in the excluded isomorphism families, as does A_7 by name.

The twenty-six sporadic groups are divided into the disjoint sets

$$\mathcal{S}_{12} = \{M_{11}, M_{12}, M_{22}, M_{23}, M_{24}, J_1, J_2, J_3, HS, He, Suz, McL\},$$
$$\mathcal{S}_{14} = \{J_4, Co_1, Co_2, Co_3, Fi_{22}, Fi_{23}, Fi'_{24}, M, B, Th, Ly, \mathrm{O'N}, HN, Ru\}.$$

For $\mathcal{S}_{12}$, the exact permutation certificates [4] check membership in each identified named group, full generation, and the unequal word orders of Lemma 4. The M_{23}, M_{24} source permutations are retained literally. Other witnesses may have different polyhedron types from the source examples; their role is existence. Named representation identifications and simplicity are retained inputs, not inferred from an order match alone.

For $\mathcal{S}_{14}$, the complete counting component [5] chooses x of an indicated odd prime order p. Counting conjugates of a maximal subgroup M containing x gives the bound

$$B_G(x) = |C_G(x)| \sum_{[M]} \frac{i_p(M) i_2(M)}{|M|}.$$

The sum includes every maximal-subgroup conjugacy class, including repeated character-table names for distinct classes. The inverter term is zero if x and x^{-1} are not conjugate in the full automorphism group; otherwise it is at most $|C_{\mathrm{Aut}(G)}(x)|\mu$. The component supplies a rigorous μ from complete maximal lists in G or its full automorphism group, checking the required centralizer indices in the four outer-index-two cases. Its fourteen strict rational inequalities prove Lemma 5. Ordinary character tables, class fusions, maximal-subgroup completeness, and the named automorphism-group identifications are explicit retained classification premises. Thus all twenty-six sporadic cases are covered without using any historical whole-map census or its numerical total.

4 The Lie families and all exceptional parameters

We use the standard classification of finite nonabelian simple groups, including its low-rank isomorphisms, to pass from these families to all simple groups. In the classical notation below, groups are projective where the centre is nontrivial. Orthogonal odd dimension has odd q; in characteristic two the corresponding simple groups are symplectic.

4.1 Finite parameter certificates

The finite component [6] supplies every simple entry in the source's two finite exception lists, together with its three exceptional Lie cases. For $\epsilon = \pm$, $L^+ = \mathrm{PSL}$ and $L^- = \mathrm{PSU}$, these are

$$L_4^\epsilon(2), L_4^\epsilon(3), L_4^\epsilon(4), L_5^\epsilon(2), L_6^\epsilon(2), \quad \mathrm{PSp}_4(3), \mathrm{Sp}_4(4), \mathrm{Sp}_6(2), \mathrm{PSp}_6(3), \mathrm{Sp}_8(2),$$
$$\Omega_7(3), \Omega_8^\epsilon(2), P\Omega_8^+(3), \Omega_{10}^\epsilon(2),$$
$$L_5^\epsilon(3), L_6^\epsilon(3), L_8^\epsilon(2), \mathrm{Sp}_6(4), \mathrm{PSp}_8(3), \mathrm{Sp}_8(4), \Omega_{12}^\epsilon(2),$$
$$G_2(3), G_2(4), {}^2F_4(2)'.$$

There are $21 + 11 + 3 = 35$ parameter entries. The source also prints $\mathrm{Sp}_4(2) \cong S_6$, which is not simple and is not a positive obligation. The overlaps $\mathrm{PSL}_4(2) \cong A_8$ and $\mathrm{PSU}_4(2) \cong \mathrm{PSp}_4(3)$ are preserved as duplicate parameter routes, not counted as new abstract groups. Each positive certificate verifies its actual named-constructor interface, membership, full generation and Lemma 4; a candidate-producing search is not the correctness premise.

4.2 Exceptional Lie type

The complete exceptional component [7] treats

$$E_8, E_7, E_6, {}^2E_6, F_4, G_2, {}^3D_4, {}^2F_4, {}^2G_2, {}^2B_2$$

in every simple parameter not covered by the finite list. The small identifications $G_2(2)' \cong \mathrm{PSU}_3(3)$ and ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$ belong to excluded families and are not converted into positive cases. The Tits group is supplied by the finite certificate, not by treating ${}^2F_4(2)$ as simple.

The component proves regularity of the chosen torus elements by a root-lattice argument, including its small finite cases. It bounds reversers by outer cosets, identifies the admissible algebraic involution classes using the reviewed flag-variety argument, and charges the cyclic normalizer. Explicit class orders give the involution lower bounds; translated polynomials with nonnegative coefficients and positive constant terms prove all uniform margins. The residual fixed parameters are checked exactly. Torus existence, complete maximal-overgroup types, class centralizers and the uniform Lawther–Liebeck–Seitz upper theorem [16] remain named antecedents. In particular, the proof does not invoke Seitz's theorem outside its field-size hypotheses or infer a finite lower constant from an asymptotic estimate.

4.3 Linear, unitary and symplectic type

The elementary classical lower component [8] proves

G	$i_2(G) > q^D/4$
$\mathrm{PSL}_n(q), \mathrm{PSU}_n(q),\ n \geq 4$	$D = \lfloor n^2/2 \rfloor$
$\mathrm{PSp}_{2m}(q),\ m \geq 2$	$D = m^2 + m$
$\Omega_{2m+1}(q),\ m \geq 3$	$D = m^2 + m$
$P\Omega^{\epsilon}_{2m}(q),\ m \geq 4$	$D = m^2 - 1.$

It counts actual involution families with their scalar losses; no maximal-subgroup classification is needed for these lower estimates. For linear and unitary groups, the complete upper component [9] constructs the Singer or $(n-1)+1$ element, proves its adjoint torus centralizer and the single possible reversing coset, and bounds the actual fixed groups, including characteristic-two nonalternating forms. It then proves the strict counting inequality for every parameter outside the displayed finite list. The exact source Table 4 linear/unitary cover is explicitly retained as a named classification assertion, with its Bereczky–Bamberg–Penttila provenance and the published author erratum. The later acquisition of an older full BP preprint does not retroactively turn that frozen component into an independent proof of the classification.

For symplectic groups, the complete component [10] uses the full Singer cover in Bubboloni–Spiga–Weigel Lemma 5.1 [17], including the odd-characteristic unitary overgroup omitted from the source table. Trace forms classify the natural reversers; the exceptional graph-field coset in degree four and even characteristic is separately included. Actual finite kernels and the degree-four estimates give strict inequalities in all remaining ranks and fields. The finite list supplies all small excluded parameters. Thus both components meet exactly the hypotheses of Lemma 5.

4.4 Orthogonal type

The torus component [11] supplies elements with their actual spinor, Dickson and projective scalar conditions. In odd dimension use a squared primitive norm-one multiplier on a negative hyperplane; in minus even dimension use a primitive multiplier for even q and its square for odd q; in plus dimension use primitive multipliers on the negative $(2m-2)+2$ decomposition. The connected projective conformal centralizer orders are respectively

$$q^m + 1, \qquad q^m + 1, \qquad (q^{m-1} + 1)(q + 1).$$

The odd and even fixed-group components [12, 13] pay the finite centralizers of natural reversing automorphisms, including nonsquare multipliers and unipotent kernels. In particular the full-rank even characteristic degree-eight bound is $\nu_2(C_G(\alpha)) \leq q^{11}$.

The orthogonal counting component [15] uses the complete BSW overgroup statements in odd dimension, minus dimension, and plus dimension at least ten. It proves the multiplicities of actual field structures, includes a bound for the cyclic normalizer, and establishes the uniform inequalities. Its remaining parameters are exactly those already present in the finite list, except for the separately treated dimension-eight range.

In plus dimension eight, the complete D4 specialization [14] proves, for every primitive choice on the negative $6+2$ decomposition, that the only maximal types are its decomposition stabilizer, at most two unitary extension normalizers, and the triality B_3 subgroups. It derives the cover from the complete published BHRD Table 8.50 [18]; the table's number of conjugacy classes is not used as an element-containing multiplicity. The independent triality argument in [15] chooses the primitive components so that all B_3 overgroups are absent for even $q \geq 4$ and odd $q \geq 7$. At $q = 5$ it counts the actual six possible line stabilizers, not the source table's asserted two. The same argument controls the additional inverter cosets. With the exact degree-eight lower estimates, the displayed strict inequalities prove the claim at all these fields. The finite certificates pay plus type at $q = 2, 3$ and minus type at $q = 2$. This is the precise interface at which the complete D4 cover discharges the conditional hypothesis of the counting manuscript; no later result is used to justify that cover.

5 Completion and source boundaries

Proof of Theorem 1. By the retained classification of finite simple groups, every group under the hypothesis is alternating, sporadic or of Lie type, with the standard small identifications described above. The alternating proof covers all eligible alternating groups. The two disjoint sporadic rosters cover all twenty-six sporadics. The finite list, exceptional component and three classical routes cover every eligible Lie parameter. Each route establishes the same full-automorphism generating-pair conditions, either by Lemma 4 or Lemma 5. There is no assumption that an auxiliary table is itself a theorem about an unverified new element. This proves the group-theoretic assertion; Lemma 2 proves Corollary 3. □

The manuscript establishes this *positive* theorem. On source page 3, nonexistence for $\mathrm{PSL}_2(q)$ is attributed to Macbeath and Singerman, and nonexistence for A_7 to the earlier atlas computation. These are historical antecedents, not deductions of our positive construction. The corresponding $\mathrm{PSL}_3(q), \mathrm{PSU}_3(q)$ statement is explicitly announced there as forthcoming work [19], not proved in the fixed source. We do not promote that announcement into a proved converse. The source's introductory classification of regular polyhedron groups, and its final almost-simple Suzuki and PSL_2-socle statements, likewise remain attributed antecedents. The general almost-simple extension is a question, not an additional conclusion.

The Section 6 counts for $M_{11}, M_{12}, M_{22}, J_1, J_2$, the cited all-types O'Nan enumeration, and example types not realized by our replacement witnesses are auxiliary historical statements. Our theorem requires their existence conclusion only. No new exhaustive map count, faithfulness of an unrelated historical census, or claim to recover every ancillary numerical example is made. In particular the O'Nan existence argument does not import the disputed total from the separate L046 source.

Several source proof interfaces have been replaced: the omitted normalizer term is covered; outer inverters are not assumed to have one class; small field hypotheses are respected; the symplectic unitary overgroup is included; orthogonal scalar losses and triality multiplicities are corrected. The source's special estimate $i_2(\mathrm{PSU}_4(q)) > q^8/2$ is false for $q \equiv 3 \pmod 4$, as established by the exact formula in [8]. The valid quarter bound and sharper finite calculations

suffice here. These are repairs to supporting assertions; they do not refute the positive principal theorem.

Attribution and retained costs. The theorem, principal strategy and alternating pairs belong to Leemans–Liebeck and their cited predecessors. The full component proofs and exact certificates listed below are part of this assembly, not merely status summaries. The classification of finite simple groups, Schulte–Weiss reconstruction, named finite-group constructions and character tables, complete maximal-subgroup classifications, stated automorphism structure, and the cited Lie-type involution inputs remain explicit borrowed results. They are not claimed to have been reproved or formally verified.

The root agent authored the alternating, twelve-sporadic, sporadic counting, exceptional, orthogonal torus, odd fixed-group, triality and orthogonal counting work. Publication_recovery authored the finite Lie certificates, classical lower bounds, linear/unitary, symplectic, even fixed-group and D4 cover components and this assembly. The root supplied the finite candidates and several organizing observations; l044_resume supplied the exceptional flag-variety amendment, which has its own separate review. The two relative-field arguments were developed in parallel and are not counted as independent reviews of each other. f003_resume supplied the bounded order-21/63 note and distinctly reviewed the general counting/multiplicity component. Source-contract assistance by l044_resume is recorded separately from mathematical acceptance. Earlier bounded proofs, their distinct reviews, and all retained classification costs remain attached to their exact editions. No novelty, new question resolution, Lean verification or new measurement is asserted.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, fixed fourteen-page source.

[2] E. Schulte and A. Ivić Weiss, *Chiral polytopes*, DIMACS Series 4 (1991), 493–516; retained rank-three reconstruction.

[3] *Alternating groups generated by a cycle and two transpositions*, complete acquired L045 alternating component, version 2, with distinct review and root adoption.

[4] *Chirality detected by the order of one word*, complete acquired twelve-sporadic component, version 2, with distinct review and adoption.

[5] *Counting involutions that generate a chiral polyhedron*, complete acquired fourteen-sporadic component, version 1, with distinct review.

[6] *Finite Lie-group witnesses for chiral polyhedra*, complete acquired thirty-five-parameter component, version 1, with distinct review.

[7] *Involutions outside the reversal and subgroup obstructions*, exceptional component, version 2, including the separately reviewed flag-variety amendment and complete arithmetic certificates.

[8] *Explicit involution classes for the classical part of L045*, version 1, with exact unitary correction and distinct review.

[9] *Torus reversers and explicit bounds for the linear and unitary branches*, version 1, with distinct review.

[10] *Explicit symplectic reverser and overgroup bounds*, version 2, with distinct review.

[11] *Orthogonal tori and their projective orders*, version 1, with distinct review.

[12] *Bounds for orthogonal fixed groups in odd characteristic*, version 2, with distinct review.

[13] *Finite kernels of full-rank orthogonal inverters in characteristic two*, version 1, with distinct root review.

[14] *Maximal overgroups of a full negative* $6 + 2$ *torus element*, version 1; complete type specialization distinctly reviewed, shared multiplicity interface separately routed through the counting review.

[15] *Counting overgroups of an orthogonal torus element* and *Choosing an orthogonal torus element to separate the triality modules*, complete orthogonal counting component, version 1; with its separate distinct review and the independently accepted complete D_4^+ cover.

[16] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific J. Math. 205 (2002), 393–464, especially Proposition 1.3.

[17] D. Bubboloni, P. Spiga and Th. Weigel, *Normal* 2*-coverings of the finite simple groups and their generalizations*, arXiv:2208.08756v1, Lemmas 5.1, 6.1, 7.3 and 8.1.

[18] J. N. Bray, D. F. Holt and C. M. Roney-Dougal, *The Maximal Subgroups of the Low-Dimensional Finite Classical Groups*, LMS Lecture Note Series 407, Cambridge University Press, 2013, Main Theorem 2.1.1 and Table 8.50.